

KineTrace

Detecting Early Motor Irregularities Using Wearable Motion Sensors and Signal Processing.

Overview

Purpose

The goal of this project is to utilize motion sensor data from everyday consumer devices to determine measurable biomechanical patterns that may correlate with early motor irregularities. More specifically, I want to use frequency-domain analysis (FDA) and smoothness metrics extracted from smartwatch IMU data in order to distinguish normal movement from tremor-like or unstable movement patterns. These methods have been used before in order to monitor people already diagnosed with disorders that signal motor dysfunction such as Parkinson's disease, however I aim to identify subtle losses in motor smoothness and stability that appear before obvious symptoms are visible by identifying micro-irregularities and hidden motion instability in human movement. Furthermore, in order to strengthen my dataset, I will use both public datasets and create my own data. Below I have included a small table to quickly summarize the key aspects of my research.

Method	Purpose
Frequency Domain Analysis	I will be using FDA in order to detect hidden oscillations/micro-tremors.
Jerk Analysis	I will analyze jerk to determine movement smoothness.
Activity Recognition	I will distinguish tremors from normal movement.
Hybrid Dataset System	I will use both public datasets and self-collected data.

Community Impact and Relevance

In addition to detecting subtle motor irregularities through wearable motion sensors, KineTrace will also examine how culture, community, and lifestyle factors may influence the risk of neurological motor diseases, particularly within elderly populations. Using the KineTrace Stability Index (KSI) which is going to be the end goal of my research, this project aims to identify early signs of reduced movement smoothness, instability, or tremor-like behavior before major symptoms become visible. My research will explore how factors such as physical activity, social interaction, participation in community programs, mobility independence, and access to healthcare may affect motor stability

over time. By combining signal processing with community health analysis, KineTrace expands beyond a technical engineering project and investigates how accessible wearable technology could support earlier detection, preventative care, and improved quality of life for aging populations.

Why Frequency-Domain Analysis?

Frequency-domain analysis or FDA is a method of analyzing signals by breaking them down into their individual frequency components rather than viewing how they change over time. FDA will be used in this project in order to analyze motion data collected from the accelerometers and gyroscopes already present inside smartwatches. Using FDA will be more useful than viewing how data changes over time for this project since irregular motor movement and tremor-like behavior often occurs as repeating oscillations at specific frequencies that are difficult to detect from simple motion graphs. In order to utilize FDA, I will be using Fast Fourier Transforms (FFTs) in order to convert signals from the time domain (amplitude vs. time) into the frequency domain (amplitude vs. frequency). This will allow me to identify dominant motion frequencies, measure oscillation intensity, and compare stable movement patterns against irregular ones quantitatively.

What is Smartwatch IMU Data?

Smartwatch inertial measurement unit (IMU) data is raw, high-frequency motion data that is collected by the smartwatch sensors. The sensors that I am going to be utilizing for this project are the accelerometer and gyroscope. These sensors measure linear acceleration and rotational movement in the three-dimensional space and continuously record motion patterns. This means I can convert movement into numerical data that can later be analyzed using frequency analysis. Since it is hard to find an application that is able to collect data from smartwatch sensors, I will be using these same sensors but on a smartphone since both use MEMS (Micro-Electro-Mechanical-Systems) technology to detect motion and orientation.

Research Question

Can smartwatch IMU data and signal-processing techniques detect subtle movement irregularities early before obvious motor dysfunction is visible?

Process

End Goal

Create a system that does the following:

1. Collects motion data from wearable sensors.
2. Analyzes movement using physics.
3. Measures motion smoothness and stability.
4. Distinguishes stable movement from irregular movement.

5. Create a web application to allow the public to use the system that is built on the algorithm created through the analysis of both collected data and public datasets
6. Create a mobile application that transforms the website into a tool that does both the data collection and analysis, further increasing accessibility.

The final result of this will be the KineTrace Stability Index (KSI) which will be a numerical score that represents movement smoothness, motion stability, and oscillatory irregularity. I believe that this score could help people detect motor dysfunction early before it becomes a significant problem and take action in order to prevent medical emergencies.

Resources

Tool	Purpose
Smartphone	Collecting data using sensors.
Sensor Logger App	Recording data and exporting it as a table.
Python	Data analysis and computations.
VS Code	Environment for programming.
NumPy	Library for computations using Python.
pandas	Library for data analysis using Python.
matplotlib	Library for creating graphs in Python.
SciPy	Library for scientific computing in Python.
scikit-learn	Library for machine learning in Python.

Data Collection

Parameter	Specification	Benchmark Alignment
Sampling Rate	50 Hz (20ms sampling window)	Matches MotionSense & UCI HAR
Sensors	3-Axis Accelerometer (ax, ay, az) + 3-Axis Gyroscope (later expanded to 6 axis with $\omega_x, \omega_y, \omega_z$).	6-Axis IMU Template
Placement	Front Right Pants Pocket (Screen facing thigh, top pointing down)	MotionSense Standard

Activity Classes	std, sit, wlk, jog, ups, dws	Standard HAR 6-Class Taxonomy
Time Per Subject	3.5 Minutes Total	Assembly Line Execution

Phase 1: Hardware & App Setup

1. Dedicated Collection Phone: Use a clean smartphone reserved exclusively for testing.
2. Download App: Install Sensor Logger (iOS / Android).
3. Lock App Settings:
 - Sample Rate: Set to 50 Hz (20ms intervals).
 - Active Sensors: Toggle ON Accelerometer and Gyroscope.
 - Disabled Sensors: Toggle OFF Magnetometer, GPS, Audio, Orientation, Barometer.
 - Output Format: CSV.
4. Physical Station Location: Identify a quiet school stairwell connected to a 50-foot flat hallway.

Phase 2: Anonymous Master Ledger

Create a Google Sheet named KineTrace_Subject_Ledger.csv. Do not record participant names. Here is an example:

id	age	sex	height_cm	weight_kg	preexisting_factors	condition_category
sub_01	17	M	178	70	None	none
sub_02	18	F	165	58	Right ACL repair (2024)	joint_injury
sub_03	17	M	172	65	Flat feet (plantar fasciitis)	gait_structural

Condition Categories:

- none: Healthy, no reported conditions or recent injuries.
- joint_injury: Ligament tears (ACL/MCL), meniscus, sprains, or dislocations.
- gait_structural: Scoliosis, flat feet, leg-length discrepancy.
- chronic_neurological: Tremors, balance disorders, inner ear issues.
- active_pain: Current muscle soreness, active limp, or minor strain.

Phase 3: The 3.5-Minute Assembly Line Protocol

1. Assign anonymous ID (e.g., sub_01).
2. Ask the standardized screening question:

3. Ask: "Do you have any current or past injuries, surgeries, joint conditions, or structural factors (like flat feet) that affect how you walk, run, or stand?"
4. Log age, gender, height_cm, weight_kg, preexisting_conditions, and condition_category in your ledger.
5. Hand the phone to the subject and instruct them: "Place this in your front right pants pocket, top of the phone pointing down, screen facing your thigh."
6. Perform the following activities:
 - Sitting (sit): Participant sits still in a chair. Press Start Recording in Sensor Logger. Wait 30 seconds. Press Stop Recording. Save export as sub_01_sit.csv.
 - Standing (std): Participant stands completely still. Press Start Recording. Wait 30 seconds. Press Stop Recording. Save export as sub_01_std.csv.
 - Walking (wlk): Participant walks down the hallway at a natural pace. Press Start Recording. Walk for 60 seconds. Press Stop Recording. Save export as sub_01_wlk.csv.
 - Jogging (jog): Participant light-jogs down the hallway or track. Press Start Recording. Jog for 30 seconds. Press Stop Recording. Save export as sub_01_jog.csv.
 - Upstairs (ups): Participant stands at the bottom of the stairwell. Press Start Recording. Participant walks up stairs for 30 seconds. Press Stop Recording. Save export as sub_01_ups.csv.
 - Downstairs (dws): Participant stands at the top of the stairwell. Press Start Recording. Participant walks down stairs for 30 seconds. Press Stop Recording. Save export as sub_01_dws.csv.

Phase 4: Nightly Batch ETL Pipeline Script

Use process_community_data.py to transform raw Sensor Logger exports into the exact kinetrace_data_template.csv schema. Then analyze the data and test it with the KineTrace web app.

Criteria

Walking

Walking is the coordinated process of moving the body forward while maintaining dynamic upright balance. The defining feature of walking is that at least one foot remains grounded at all times. Each full stride consists of two continuous phases:

- Stance Phase: The period where the foot is in contact with the ground, progressing smoothly from initial heel contact, through mid-foot support, to toe push-off.
- Swing Phase: The period where the foot lifts off the ground, swings forward through the air, and prepares for the next contact.

Observable Quantities

- Forward Pace & Tempo: Assessing overall walking flow and rhythm.
- Bilateral Balance: Comparing left side motion against right side motion.
- Torso Stability: Observing side-to-side sway and upright control.
- Overall Stability: Synthesizing movement smoothness and repeatability into a general health metric.

Good vs. Unstable

Aspect	Positive	Negative
Limb Symmetry	Left and right steps mirror each other in timing, stride length, and weight bearing.	Limping, favoring one leg, or noticeable differences between left and right steps.
Rhythm	Steady, predictable, and metronomic pace from step to step.	Irregular, unpredictable pacing, sudden hesitations, or stumbling motion.
Torso Control	Upright spine with gentle, fluid vertical motion and minimal side-to-side wobble.	Excessive side-to-side torso sway, forward slumping, or rigid body compensation.
Foot Mechanics	Clean heel landing rolling smoothly to push-off, with ample foot clearance during swing.	Toe dragging, foot slapping, or landing flat-footed on the ground.

Fall Indicators

- Shuffling: Short, rapid, low-clearance steps where the feet barely lift off the floor.
- Antalgic Gait: A rapid, hurried step off an uncomfortable or painful leg to limit weight bearing.
- Ataxic Gait: An unsteady, wide-based, wobbling movement pattern indicating balance or coordination difficulty.
- Pacing Instability: Inability to maintain a continuous rhythm, leading to sudden tripping or losing balance.

Evaluation

1. The 3 Planes of Motion
 - Forward-Backward (Flow): Look for continuous, steady momentum versus sudden jerks, surges, or hesitation spikes.
 - Side-to-Side (Balance): Look for tight weight transfer directly over the landing foot versus excessive torso swaying.

- Vertical (Clearance & Impact): Look for fluid bouncing and clean foot lifting versus harsh landing impacts or foot dragging.
2. Left vs. Right Symmetry
 - Time: Is stance duration identical on both legs?
 - Impact: Is heel-strike force equal on both sides?
 - Push-off: Is forward propulsion equal from both feet?
 3. Signal Smoothness & Repeatability
 - Rhythm: Does the movement repeat as a smooth, predictable wave step after step?
 - Transitions: Are handoffs between foot-on-ground and foot-in-air seamless and stumble-free?
 - Pacing: Does the rhythm stay steady without unexplained speed changes?

Reference Papers

1. 2D Video Pose Estimation for Spatiotemporal Gait Analysis
 - Paper Link: <https://pmc.ncbi.nlm.nih.gov/articles/PMC8099131/>
 - Methodology: Compares OpenPose keypoint tracking against 3D motion capture during overground walking to measure step time, stance time, swing time, and step length.
 - Dataset: Overground walking video recordings from healthy adults across multiple walking speeds.
 - How It Helps Our Goal: Proves that standard camera video can accurately quantify Limb Symmetry (left vs. right stance/swing duration) and Rhythm (step time consistency) down to a 0.01-second margin of error, confirming we do not need pressure plates to measure step timing.
2. Computer Vision & Temporal Gait Feature Extraction for Fall Risk
 - Paper Link: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11389313/>
 - Methodology: Extracts temporal parameters (cadence, step time, stride time, and stance symmetry) from 30fps video during Timed Up and Go (TUG) assessments using pose estimation to classify fallers versus non-fallers.
 - Dataset: MMU-FRiP (Multimedia University Fall Risk Prediction) dataset.
 - How It Helps Our Goal: Directly aligns with our Fall Indicators (shuffling, pacing instability) and Bilateral Balance, demonstrating how simple keypoint trajectories can identify fall risk automatically.
3. Large-Scale Clinical Gait Abnormality Dataset (GAVD)
 - Paper Link: <https://arxiv.org/abs/2407.04190>
 - Methodology: Uses deep video action recognition models (Temporal Segment Networks / SlowFast) on RGB video sequences to detect pathological walking patterns in unconstrained environments.

- Dataset: The GAVD dataset, containing 1,874 video sequences across 400+ subjects exhibiting normal, abnormal, and pathological gaits (e.g., shuffling/festinating gait, limping).
 - How It Helps Our Goal: Provides a large, benchmark dataset to train and test vision algorithms specifically for detecting qualitative gait abnormalities from our outline, such as Ataxic Gait, Antalgic Gait, and Torso Control (excessive sway/slumping).
4. Single Lower-Back IMU Spatiotemporal Gait Analysis
- Paper Link: <https://pmc.ncbi.nlm.nih.gov/articles/PMC11389313/>
 - Methodology: Tracks initial foot contact (IC), final contact (FC), cadence, and step time variability using a single inertial sensor placed on the lower back over the sacrum.
 - Dataset: Clinical mobility datasets capturing continuous overground walking across fallers and non-fallers.
 - How It Helps Our Goal: Mirrors our exact hardware setup (a single lower-back accelerometer) and demonstrates how to process a_x , a_y , and a_z signals to detect heel strikes, step symmetry, and smoothness without pressure plates.

Sitting

Sitting transitions (sitting down and standing up) involve controlled vertical shifts in center of mass, transferring body weight between the feet and a supporting surface. Each transition consists of two continuous phases:

- Trunk Flexion Phase: The period where the upper body leans forward at the hips to position the center of mass over the feet while knees begin bending.
- Vertical Transfer Phase: The period of controlled lowering onto a seat (eccentric control) or driven extension upward to full upright standing (concentric power).

Observable Quantities

- Transition Tempo & Duration: Assessing overall speed, momentum, and fluidity when changing position to evaluate dynamic coordination, pace management, and seamless movement efficiency.
- Bilateral Load Transfer: Comparing left-leg versus right-leg weight distribution during elevation and lowering to evaluate movement symmetry, dynamic balance control, and subtle compensatory force shifts.
- Trunk Sway & Control: Observing forward lean angle and lateral torso wobble during ascent or descent.
- Overall Stability: Synthesizing smooth velocity curves and control into a general mobility score that quantifies functional balance and biomechanical efficiency.

Good vs. Unstable

Aspect	Positive	Negative
Limb Symmetry	Left and right legs apply equal force and flex evenly during sitting down or standing up	Pushing off or collapsing heavily onto one side, or favoring one leg during the transition.
Rhythm	Fluid, continuous transition without stopping mid-movement.	Hesitation, mid-motion freezes, or multi-attempt rocking to generate momentum.
Torso Control	Controlled forward trunk lean with stable lateral spinal alignment.	Excessive side-to-side torso wobble, forward slumping, or uncontrolled backwards tipping.
Foot Mechanics	Smooth, continuous vertical acceleration landing softly on the seat or rising evenly.	"Plopping" or collapsing onto the seat; uncontrolled falling trajectory due to poor eccentric control.

Fall Indicators

- Plop Sitting: Sudden loss of eccentric control causing a hard crash onto the seat.
- Asymmetric Drop: Leaning or collapsing toward one side during lowering or standing push-off.
- Multi-Attempt Rocking: Inability to rise in a single continuous movement, requiring repeated forward momentum rocks.
- Post-Transition Instability: Excessive torso wobble or forward stumbling immediately upon standing up.

Evaluation

1. Vertical Power & Eccentric Control
 - Lowering Acceleration: Look for smooth, continuous vertical deceleration into the seat versus an uncontrolled falling trajectory or sudden collapse.
 - Rising Thrust: Look for strong, continuous upward acceleration to lift body mass versus struggling, hesitation, or mid-movement freezing.
 - Landing Impact: Look for soft, damped contact at seat touch-down versus harsh vertical jerk spikes (plopping).
2. Trunk Flexion & Pitch Kinematics
 - Forward Lean Flow: Look for clean forward upper-body rotation to shift center of mass over feet before lift-off versus insufficient or jerky leaning.

- Lateral Wobble: Look for centered spinal alignment versus excessive side-to-side torso sway during elevation change.
 - Transition Fluidity: Look for seamless handoffs between trunk flexion and vertical displacement without momentum loss or multiple rocking attempts.
3. Bilateral Load & Timing Symmetry
- Force Distribution: Is weight-bearing force applied equally through both legs during standing push-off and seat landing?
 - Phase Duration: Is the duration of leg flex and extension identical on both sides during ascent and descent?
 - Propulsion Balance: Is upward propulsion generated evenly from both hips without rotational twisting or leaning heavily over one limb?

Reference Papers

1. Metrics Extracted from a Single Wearable Sensor During Sit-to-Stand and Stand-to-Sit
 - Paper Link: <https://pmc.ncbi.nlm.nih.gov/articles/PMC7413823/>
 - Methodology: Placed a single inertial measurement unit (IMU) on the lower back (L4–L5 sacral region) to extract trunk flexion velocity, vertical acceleration peaks, transition duration, and jerk values during sit-to-stand and stand-to-sit cycles.
 - Dataset: Clinical cohort performing 5-Times Sit-to-Stand (5STS) tests across varying functional mobility levels.
 - How It Helps Our Goal: Directly validates your hardware location (lower back) and proves that 3D accelerometer signals (ax,ay,az) can isolate trunk flexion from vertical elevation, allowing algorithms to detect smooth velocity curves versus landing impact spikes ("plopping").
2. Validation of a Lower Back Wearable-Based Sit-to-Stand and Stand-to-Sit Transition Detection Algorithm
 - Paper Link: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6104484/>
 - Methodology: Developed heuristic and signal-processing algorithms to detect postural transition start/end points, duration, and peak power using a single waist/lower-back accelerometer.
 - Dataset: Wearable sensor telemetry recorded in semi-unstructured environments across healthy and mobility-impaired individuals.
 - How It Helps Our Goal: Informed your algorithm structure for classifying transition phases (Flexion vs. Vertical Transfer), showing how raw accelerometer signals can measure transition fluidity and multi-attempt rocking without physical lab equipment.
3. Quantitative Assessment of the 30-Second Sit-to-Stand Test Using Video Pose Estimation
 - Paper Link: <https://pmc.ncbi.nlm.nih.gov/articles/PMC13326458/>

- Methodology: Applied keypoint tracking (hip, knee, ankle, shoulder keypoints via Keypoint R-CNN) on RGB video streams to extract trunk tilt angles, angular velocity, and vertical displacement profiles during sit-to-stand transitions.
- Dataset: Monocular 2D video recordings of subjects performing standard chair-rise assessments.
- How It Helps Our Goal: Demonstrates that standard video camera footage can measure trunk lean angles and bilateral knee/hip extension symmetry, proving computer vision keypoints match lab-grade motion capture for sitting mechanics.

Standing

Maintaining an upright, static posture with feet planted on a supporting surface, continuously controlling the body's Center of Mass (CoM) to remain within the Base of Support (BoS).

- Ankle Strategy Phase (Low-Amplitude Correction): Primary inverted-pendulum control where subtle CoM drifts are continuously corrected through micro-torques generated by ankle plantarflexors/dorsiflexors (anteroposterior) and invertors/evertors (mediolateral).
- Hip Strategy Phase (High-Amplitude / Multi-Joint Correction): Recruitment of trunk, hip, and upper-body counter-rotations when perturbations exceed ankle torque capabilities or when stance width is narrowed (e.g., tandem or single-leg stance).

Observable Quantities

- AP & ML Sway Amplitude / Acceleration RMS: Root mean square acceleration of the lower back (L4–L5) or spatial CoM trajectory along the anteroposterior and mediolateral axes (mm or m/s^2).
- Mean Sway Velocity / Path Length: Cumulative spatial distance traversed by the CoM per unit time (mm/s), reflecting overall energetic effort spent maintaining static balance.
- 95% Confidence Ellipse Area: Surface area (cm^2) enclosing 95% of the total CoM trajectory over a fixed duration (e.g., 30 seconds).
- Dominant Sway Frequency & Spectral Power: Fast-Fourier Transform (FFT) breakdown categorizing low-frequency drift (<0.5 Hz) versus high-frequency corrections (>1 Hz).

Good vs. Unstable

Aspect	Positive	Negative
Postural Sway & Amplitude	Minimal, tight AP/ML displacement enclosed	Excessive, erratic spatial drift with an expanded

	within a small 95% confidence ellipse area ($<5\text{cm}^2$).	sway area ($>15^2$) or severe directional asymmetry.
Joint Strategy & Rigidity	Smooth, fluid ankle-dominant micro-corrections with relaxed knees and pelvis.	Over-reliance on hip sway, rigid co-contraction of trunk muscles, or visible postural tremor.
Base of Support & Foot Placement	Equal bilateral weight distribution with stable stance width and full foot contact.	Involuntary weight shifting, foot lifting, heel taps, or compensatory widening of stance.
Sensory Compensation	Minimal change in sway magnitude when closing eyes or standing on soft surfaces.	Sudden loss of balance, severe sway expansion, or immediate stepping reaction upon closing eyes (Romberg sign).

Fall Indicators

- Rapid Sway Velocity Spikes: Abrupt bursts in CoM velocity ($>30\text{ mm/s}$) indicating delayed or failing sensory feedback loops.
- Directional ML Instability: Lateral (side-to-side) sway magnitude exceeding anteroposterior sway, strongly associated with lateral hip impacts in older adults.
- Compensatory Stepping Reactions: Unplanned foot repositioning, stumbles, or boundary steps during static stance testing.
- High-Frequency Postural Tremor: Rapid micro-oscillations ($>2\text{ Hz}$) reflecting motor unit firing dysfunction or extreme muscular fatigue.

Evaluation

1. Spatial Sway & Displacements
 - AP & ML Trajectory Range: Look for tight, controlled center-of-mass boundaries versus excessive anteroposterior or lateral excursion indicating impaired hip control and fall risk.
 - 95% Confidence Ellipse Area: Look for a compact surface area enclosing total body sway versus an expanded footprint during eyes-closed or perturbed stance.
 - Center-of-Mass Spatial Drift: Look for centered body alignment relative to the base of support versus persistent leaning or asymmetrical weight distribution.
2. Kinematic Velocity & Frequency Dynamics
 - Sway Velocity Spikes: Look for low, consistent speed of body displacement versus sudden velocity surges revealing reactive rescue adjustments after micro-slips.

- Jerk & Acceleration Smoothness: Look for fluid, automated postural corrections (low jerk) versus erratic, uncoordinated muscular co-contractions.
 - Spectral Frequency Bands: Look for dominant low-frequency sensory adjustments versus high-frequency oscillations reflecting muscle stiffness, anxiety, or neurological tremor.
3. Stance Strategy & Sensory Dependence
- Ankle vs. Hip Strategy: Look for fine, ankle-driven micro-corrections (normal inverted-pendulum control) versus early reliance on hip sway and upper-body counter-rotations.
 - Visual Dependence (Romberg Ratio): Look for stable balance maintenance without visual cues versus a dramatic expansion in sway when eyes are closed.
 - Base of Support Adaptation: Look for steady, continuous foot contact versus involuntary weight shifts, heel lifting, or compensatory stepping reactions.

Reference Papers

1. Wearable Inertial Sensors to Assess Standing Balance
 - Paper Link: <https://pmc.ncbi.nlm.nih.gov/articles/PMC6806601/>
 - Methodology: Analyzes lower-back (L4–L5) accelerometer/IMU positioning and signal processing algorithms across static balance protocols (eyes open/closed, firm/foam surfaces).
 - Dataset: Synthesized findings across 50+ clinical studies encompassing healthy adults, Parkinson's disease, concussion, and fall-risk populations.
 - How It Helps Our Goal: Validates standard sacral IMU placement and provides signal processing formulas to extract acceleration RMS, sway velocity, and spectral frequency metrics during quiet stance.
2. Accelerometry Applications and Methods to Assess Standing Balance
 - Paper Link: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10881067/>
 - Methodology: Validates lumbar 3D accelerometry against laboratory force-plate Center of Pressure (CoP) gold standards during static quiet stance tasks.
 - Dataset: Multi-cohort clinical dataset across healthy young/elderly adults and neurological patients.
 - How It Helps Our Goal: Proves that lower-back IMU acceleration signals strongly correlate ($r > 0.85$) with laboratory force-plate sway metrics (trajectory range, 95% confidence ellipse area), confirming force plates aren't necessary.
3. Markerless Pose Estimation for Standing Balance Level Recognition
 - Paper Link: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10803458/>

- Methodology: Applies markerless pose estimation keypoints from RGB video streams to calculate center-of-mass kinematics, joint angles, and postural sway path lengths during static standing tests.
- Dataset: Video recordings of healthy adults and patients with musculoskeletal balance disorders.
- How It Helps Our Goal: Demonstrates that standard video camera pose tracking can accurately monitor static postural sway and trunk tilt angles without marker-based motion capture.

Walking Up Stairs

Walking up stairs (stair ascent) is the dynamic vertical and forward elevation of the body's center of mass against gravity over discrete steps. Stair climbing demands significantly higher lower-limb joint moments and concentric muscular power than level walking. Each full stair step cycle consists of two continuous phases:

- Stance / Pull-Up Phase: The period initiated by foot contact on the step tread, followed by lower-limb extension (quadriceps and gluteals) to elevate body mass vertically and drive the body forward over the lead leg.
- Forward Continuance / Swing Phase: The period where the trailing foot unloads, lifts off the lower step, and swings forward and upward to clear the step riser and establish placement on the next tread.

Observable Quantities

- Vertical Thrust & Flow: Assessing upward propulsion force, acceleration smoothness, and continuous elevation momentum.
- Stepping Rhythm & Cadence: Measuring step-to-step timing consistency, climbing tempo, and flight completion time.
- Bilateral Push-off Balance: Comparing left-leg versus right-leg concentric extension power and single-limb stance stability.
- Torso Pitch & Clearance: Observing forward trunk lean angle, lateral sway containment, and foot-to-riser clearance height.

Good vs. Unstable

Aspect	Positive	Negative
Limb Symmetry	Both legs generate equal upward thrust and share weight bearing evenly across step transitions.	Favoring one leg, trailing-leg dragging, or asymmetrical single-leg pulling during elevation.
Rhythm	Continuous, metronomic climbing pace with predictable cadence per stair riser.	Irregular step timing, mid-flight pauses, or multi-attempt foot placements on a single

		tread.
Torso Control	Controlled forward trunk lean centered over the base of support with stable spinal alignment.	Excessive forward slumping, side-to-side torso wobble, or heavy reliance on momentum/handrails.
Step Clearance	Ample foot clearance over each stair riser with clean, centered foot landing on the tread.	Low clearance, toe scuffing/catching against risers, or landing flat on the tread edge.

Fall Indicators

- Toe Catching / Scuffing: Insufficient foot elevation during the swing phase, causing the toe to strike the stair riser and risking a forward trip.
- Hesitation Freezing: Sudden loss of vertical momentum or pausing mid-step before initiating the concentric pull-up phase.
- Asymmetric Vaulting / Leaning: Leaning heavily to one side or using upper-body swinging to compensate for unilateral lower-limb extension weakness.
- Non-Reciprocal Climbing: Inability to alternate feet from step to step (step-together gait), placing both feet on a single tread before advancing.

Evaluation

1. Vertical Propulsion & Acceleration Flow
 - Upward Elevation Smoothness: Look for fluid, continuous vertical acceleration (a_z) during the concentric pull-up phase versus jerky, hesitant acceleration spikes.
 - Concentric Thrust Power: Look for decisive upward force production per step versus sluggish, prolonged extension effort.
 - Momentum Continuity: Look for uninterrupted vertical transition from step to step versus complete momentum loss between risers.
2. Bilateral Power & Ascent Symmetry
 - Limb Extension Balance: Look for equal vertical acceleration peak heights generated by left and right lead legs versus marked side-to-side thrust asymmetry.
 - Lateral Sway Containment: Look for minimal mediolateral trunk sway (a_y) during single-leg support on the step tread versus heavy lateral leaning over the stance foot.
 - Phase Duration Equality: Look for identical stance-to-swing time ratios for both left and right legs during vertical advancement versus unequal single-leg support times.
3. Cadence & Step Clearance Dynamics

- Climbing Cadence & Rhythm: Look for steady step frequency (dominant FFT peak) throughout the stair flight versus erratic, unpredictable step timing.
- Riser Clearance Margin: Look for clean vertical foot trajectories swinging over step edges versus low-margin trajectory curves risking toe impacts.
- Post-Ascent Stabilization: Look for a smooth, immediate transition back to level overground walking upon reaching the top landing versus forward stumbling or post-ascent sway spikes.

Reference Papers

1. Wearable-Based Stair Climb Power Estimation and Activity Classification
 - Paper Link: <https://pmc.ncbi.nlm.nih.gov/articles/PMC9459813/>
 - Methodology: Uses a single lumbar-worn tri-axial accelerometer to classify stair ascent periods vs. descent and level walking, calculating step power output (W/kg) and signal intensity.
 - Dataset: Subjects performing stair negotiation protocols with wearable sensors.
 - How It Helps Our Goal: Validates using a single lower-back accelerometer to detect stair climbing cycles, isolate step transitions, and calculate concentric vertical thrust and step power.
2. Older Adults' Lower-Limb Muscle Power Production Throughout a Full Flight of Stairs
 - Paper Link: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10868844/>
 - Methodology: Evaluates step-by-step acceleration, initial acceleration phase (P1), and peak power (Pmax) during stair climbing using a single lower-back body-fixed accelerometer (L4/L5 region).
 - Dataset: Older and young adult cohorts performing full-flight stair ascent.
 - How It Helps Our Goal: Demonstrates how L4/L5 IMU signals can quantify vertical acceleration smoothness, peak concentric power, and step cadence during stair ascent.
3. Children With Developmental Coordination Disorder Show Altered Kinematics and Task Optimization Strategies During Stair Negotiation
 - Paper Link: <https://pmc.ncbi.nlm.nih.gov/articles/PMC7731582/>
 - Methodology: Analyzes 2D video spatial-temporal parameters (vertical toe clearance, step duration, trajectory variability) across different zones of a staircase during ascent.
 - Dataset: Video recordings of participants during stair negotiation tasks.
 - How It Helps Our Goal: Proves that computer vision and video analysis can accurately track vertical toe-to-riser clearance and trajectory consistency to identify fall risks.

Walking Down Stairs

Walking down stairs (stair descent) is the controlled downward and forward movement of the body's center of mass against gravity over discrete steps. Stair descent requires high eccentric muscle control and generates substantially higher impact loading forces than level walking or stair ascent, making it one of the highest-risk daily activities for falls.

Each full stair descent cycle consists of two continuous phases:

- **Weight Acceptance / Controlled Lowering Phase:** The period initiated by foot contact on the lower step tread, followed by controlled eccentric knee and ankle flexion of the stance leg to absorb impact forces and lower the body mass smoothly.
- **Forward Swing / Stepping Phase:** The period where the trailing leg unloads, lifts off the upper step, and swings forward and downward over the step nosing to establish placement on the next lower tread.

Observable Quantities

- **Impact Damping & Control:** Assessing vertical force absorption, land-impact smoothness, and eccentric lowering control.
- **Stepping Rhythm & Cadence:** Measuring step-to-step timing consistency, lowering tempo, and flight completion time.
- **Bilateral Load Transfer:** Comparing left-leg versus right-leg eccentric lowering stability and single-limb support times.
- **Torso Pitch & Landing Alignment:** Observing backward lean or forward tipping, side-to-side torso sway, and foot-to-step alignment.

Good vs. Unstable

Aspect	Positive	Negative
Limb Symmetry	Both legs absorb landing forces equally and exhibit symmetrical knee flexion during lowering.	Favoring one leg, stiff-legged landings, or dropping heavily onto one side.
Rhythm	Continuous, metronomic downward pace with predictable cadence across steps.	Irregular step timing, acceleration surges (controlled falling), or mid-flight hesitation.
Torso Control	Erect spine with slight forward tilt centered over the landing foot and stable lateral alignment.	Excessive backward leaning (fear reaction), forward slumping, or lateral torso sway.
Step Clearance	Clean midfoot/forefoot touchdown centered on the	Heel catching on step edges, foot slapping, or

tread with ample heel clearance over the nosing	landing precariously on the tread border.
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Fall Indicators

- Uncontrolled Descent (Controlled Falling): Inability to decelerate downward momentum, resulting in progressive speeding up down the stair flight.
- Foot Slapping / Drop Landing: Sudden loss of eccentric quadriceps/plantarflexor control, causing a hard impact crash on the lower step tread.
- Heel Catching / Edge Miss: Insufficient forward clearance causing the trailing heel to strike the stair nosing or landing with foot overhanging the tread edge.
- Post-Descent Instability: Inability to smoothly arrest downward momentum upon reaching the bottom landing, resulting in forward stumbling.

Evaluation

1. Impact Damping & Eccentric Control
 - Look for smooth vertical acceleration deceleration (a_z) at foot contact versus sharp, high-magnitude impact spikes (foot slap or drop landing).
 - Look for continuous downward momentum absorption during knee flexion versus rigid, jarring joint landings or abrupt momentum freezes.
 - Look for low vertical jerk values during weight transfer, indicating controlled eccentric quadriceps and calf muscle activation.
2. Bilateral Load & Descent Symmetry
 - Look for equal impact force magnitudes and deceleration curves across both left and right landing legs.
 - Look for minimal mediolateral trunk sway (a_y) during single-limb lowering on the upper tread versus heavy lateral tilt over the stance side.
 - Look for symmetrical stance-to-swing phase durations on left and right legs during downward progression.
3. Cadence & Foot Placement Dynamics
 - Look for a steady, metronomic downward stepping frequency (dominant FFT peak) versus chaotic, unpredictable step timing.
 - Look for clean trajectory clearance of the swinging foot over the step nosing edge versus low-margin curves risking heel trips.
 - Look for a seamless transition from the final step onto level overground walking without forward stumbling or post-descent sway surges.

Reference Papers

1. Comparison of Trunk Acceleration Ratios During Stair Negotiation in Old-Old Females
 - Paper Link: <https://pmc.ncbi.nlm.nih.gov/articles/PMC4932088/>

- Methodology: Directly measures lower-back (L3) and upper-trunk (T2) tri-axial accelerations during stair descent to evaluate acceleration ratios, high-frequency impact vectors, and postural instability.
 - Dataset: Female cohorts undergoing stair negotiation assessments.
 - How It Helps Our Goal: Validates lower-back accelerometry for detecting high vertical impact loading, trunk acceleration spikes, and eccentric instability during stair descent.
2. A Novel Smart Shoe Instrumented with Sensors for Quantifying Foot Placement and Clearance During Stair Negotiation
 - Paper Link: <https://pmc.ncbi.nlm.nih.gov/articles/PMC10747814/>
 - Methodology: Quantifies toe-to-step edge clearance and foot placement variability during stair descent to assess trip and slip risk.
 - Dataset: Participants navigating stairs with foot placement tracking.
 - How It Helps Our Goal: Establishes foot-to-step edge clearance as the primary metric for stair descent stability, guiding our video pose tracking models for step-edge margin analysis.

Algorithm

System Overview & Data Flow Pipeline

The KineTrace algorithm transforms raw 6-axis IMU sensor telemetry (3-axis accelerometer a_x, a_y, a_z and 3-axis gyroscope $\omega_x, \omega_y, \omega_z$) recorded at 50 Hz into a clinical stability score called the KineTrace Stability Index (KSI). The Python pipeline executes sequentially through six distinct stages across the codebase:

1. Data Harmonization & ETL (download_motionsense.py, download_uci.py, process_community_data.py, harmonize_data.py, build_dataset.py, build_motionsense_dataset.py)
 - Ingests public benchmarks (MotionSense, UCI HAR) and self-collected community CSVs, aligning them into a standardized 50 Hz master matrix.
2. Dynamic Signal Preprocessing & Multi-Filter Conditioning (extract_features.py)
 - Computes magnitude vectors ($|a(t)|$, $|\omega(t)|$) to eliminate phone/watch orientation dependence.
 - Applies user-selected dynamic digital filters (IIR, FIR, Adaptive, or Spatial/Nonlinear) to isolate movement components and suppress noise.
3. Feature Extraction & Signal Processing (extract_features.py, explore_signals.py)
 - Extracts time-domain metrics (Dimensionless Jerk, RMS, 95% Sway Ellipse Area).
 - Runs Fast Fourier Transforms (FFT) to extract spectral energy distribution and 3-8 Hz tremor power ratios.
4. Machine Learning & Fall Risk Classification (train_universal_model.py, calculate_fall_risk.py)

- Applies a Random Forest classifier to recognize current activity (HAR) and output multi-class fall risk probabilities.
5. Task Evaluation & KSI Engine (calculate_stability.py)
 - Evaluates task-specific biomechanics across walking, sitting, standing, stair ascent, and stair descent.
 - Synthesizes all extracted metrics into the unified 0-100 KineTrace Stability Index (KSI).
 6. Application Serving & Real-Time Telemetry (main.py, live_socket_server.py, run_kinetrace_analysis.py)
 - Delivers REST endpoints (FastAPI) and WebSocket channels for web app integration, live data streaming, and batch CLI processing.

Dynamic Signal Conditioning & Exact Filter Selection Engine

To ensure measurements remain invariant to device placement or pocket angle, all tri-axial raw signals are first converted into Euclidean vector magnitudes inside `extract_features.py`:

$$|a(t)| = \sqrt{a_x(t)^2 + a_y(t)^2 + a_z(t)^2}$$

$$|w(t)| = \sqrt{w_x(t)^2 + w_y(t)^2 + w_z(t)^2}$$

Dynamic Filter Options

Users can select from eight dynamic filtering algorithms within the web analyzer interface. When a configuration request reaches `main.py`, the parameter schema routes raw telemetry directly to the designated mathematical filter in `extract_features.py`:

1. Butterworth Lowpass (Butterworth lowpass): A 4th order maximally flat passband low-pass filter $f_c=15$ Hz that eliminates high-frequency electronic jitter and sensor noise while fully preserving voluntary human motor control dynamics (0.5-10 Hz).
2. Chebyshev Highpass (Chebyshev highpass): A Chebyshev Type I high-pass filter ($f_c=0.25$ Hz) featuring a steep transition roll-off rate. Strips the static 1g gravitational acceleration baseline to isolate pure dynamic linear acceleration.
3. Bessel Bandpass (Bessel bandpass): A linear-phase band-pass filter configured to the 3-8 Hz frequency window. Preserves transient pulse wave shapes without phase distortion to accurately isolate involuntary neurological micro-tremors.
4. Median Filter (Median filter): A non-linear rank-order filter that replaces each sample with the median of neighboring data points within a sliding frame window. Strips high-amplitude single-sample spikes, packet drops, and hardware sensor glitches.

5. Gaussian Smooth (Gaussian smooth): Convolves time-series data with a 1D Gaussian kernel distribution. Provides smooth attenuation of rapid micro-fluctuations while preserving macroscopic movement trends.
6. Kalman Filter (Kalman filter): A recursive linear quadratic state estimator that models true body acceleration states against sensor measurement variance, producing optimal real-time state estimates under Gaussian noise.
7. Savitzky-Golay (Savitzky-Golay): A local least-squares polynomial smoothing filter applied across sliding window frames. Attenuates high-frequency noise while preserving peak height, peak width, and transient acceleration dynamics during high-impact transitions.
8. Wavelet Denoise (Wavelet denoise): Uses discrete wavelet decomposition (e.g., Daubechies / Symlet basis wavelets) with soft-thresholding to decouple non-stationary biomechanical transient events from background noise across multiple frequency scales.

Mathematical Feature Extraction

Following signal conditioning, `extract_features.py` calculates time-domain and frequency-domain markers of motor instability:

1. Dimensionless Jerk (Movement Smoothness)

Jerk represents the rate of change of acceleration over time (da/dt). Lower jerk values correspond to fluid, coordinated movement, whereas high jerk spikes indicate micro-instabilities, muscle corrections, or uncoordinated motor control:

$$J_{norm} = \sqrt{\frac{1}{2T} \int_0^T \left(\frac{d|a(t)|}{dt} \right)^2 dt} \times \frac{T^3}{v_{mean}}$$

2. Fast Fourier Transform & Spectral Tremor Ratio

Signal windows are converted into frequency spectra using Discrete Fourier Transforms:

$$X(f) = \sum_{n=0}^{N-1} |a[n]| \times e^{-\frac{i2\pi fn}{N}}$$

To quantify sub-clinical motor irregularities, the Spectral Tremor Power Ratio (Rtremor) measures energy density within the pathological tremor band (3-8 Hz) relative to voluntary movement energy (0.5-3 Hz):

$$R_{tremor} = \frac{\int_3^8 PSD(f) df}{\int_{0.5}^3 PSD(f) df}$$

3. 95% Confidence Postural Sway Ellipse Area

During static standing tasks, center-of-mass motion along the anteroposterior (x) and mediolateral (y) axes is bounded by a spatial confidence ellipse:

$$Area_{95\%} = 5.991 \times \pi \times \sigma_x \sigma_y \sqrt{1 - \rho_{xy}^2}$$

where σ_x and σ_y are the standard deviations of sway displacement, and ρ_{xy} is the cross-correlation coefficient.

Universal Machine Learning & Fall Risk Engine

1. Dataset Harmonization (harmonize_data.py, build_dataset.py)
Public benchmark datasets (MotionSense and UCI HAR) are ingested via download_motionsense.py and download_uci.py. Self-collected field data (processed through process_community_data.py) is merged into a single 50 Hz master matrix containing six activity classes: Sitting (sit), Standing (std), Walking (wlk), Jogging (jog), Stair Ascent (ups), and Stair Descent (dws).
2. Universal Model Training (train_universal_model.py) A Random Forest Classifier ensemble (100 decision trees) is trained on the harmonized feature matrix. The model uses multi-class probability outputs to accurately identify the active movement state and flag anomalous activity signatures.
3. Fall Risk Assessment (calculate_fall_risk.py) calculate_fall_risk.py combines the Random Forest probability distributions (P risk) with heuristic clinical risk markers (e.g., sudden velocity spikes >30 mm/s, stance asymmetry >15%, landing impact jerks >35m/s³).

Task-Specific Evaluation & The KineTrace Stability Index

calculate_stability.py performs targeted biomechanical evaluations based on the active movement domain, then synthesizes all metrics into the final KineTrace Stability Index (KSI).

Task Evaluation Criteria

Movement Task	Primary Evaluated Features	Target Normal Benchmark	Pathological / Instability Indicator
Walking	Step time symmetry, cadence variance	Asymmetry < 5%	Asymmetry > 15% (Ataxic / Antalgic gait)
Sitting	Eccentric lowering impact jerk, transition flow	Impact Jerk < 12 m/s ³	Uncontrolled collapse / "Popping" >35 m/s ³)

Standing	95% Sway Ellipse Area, Tremor Power Ratio	Area < 5.0 cm ²	Area > 15.0 cm ² or R _{tremor} > 0.35
Stair Ascent	Concentric thrust power, step rhythm consistency	Smooth vertical thrust	Hesitation freezing / Asymmetric vaulting
Stair Descent	Eccentric impact damping ratio, cadence stability	Damping Ratio < 1.8	Harsh foot slap / Uncontrolled deceleration

KineTrace Stability Index (KSI) Mathematical Formula

$$KSI = 100 - \left[w_1 \left(\frac{J_{norm}}{J_{max}} \right) + w_2 \left(\frac{R_{tremor}}{R_{max}} \right) + w_3 S_{asymmetry} + w_4 P_{risk} \right] \times 100$$

Where $w_1 = 0.30$, $w_2 = 0.25$, $w_3 = 0.25$, and $w_4 = 0.20$.

Clinical Score Bands

- 90 – 100 (Optimal Motor Control): High fluidity, low dimensionless jerk, negligible 3-8 Hz spectral oscillations, symmetrical stride mechanics.
- 75 – 89 (Mild Functional Variation): Intact postural control with minor movement jerk spikes during elevated effort transitions.
- 55 – 74 (Sub-Clinical Micro-Irregularity): Early losses in movement smoothness, elevated tremor-band energy ratio, or expanded standing sway area (>10cm²). Flagged for longitudinal tracking.
- < 55 (Elevated Instability / High Fall Risk): Severe gait asymmetry (>15%), uncontrolled transition impact force, or high Random Forest fall risk probability P risk > 0.70). Recommends clinical evaluation.

Execution & Deployment Architecture

- main.py: Production FastAPI application providing REST endpoints (/api/analyze, /api/filter-config) that receive client telemetry, execute filtering and feature extraction, run model inference, and return JSON stability metrics.
- live_socket_server.py: WebSocket server enabling real-time, low-latency streaming of wearable telemetry for live spectral and score visualization.
- run_kinetrace_analysis.py: Command-line pipeline execution script for batch-processing subject files end-to-end.
- simulate_client.py & test_api.py: Automated testing framework for verifying API latency, WebSocket data packet handling, and model inference speed.
- verify_setup.py: System validation script ensuring dependencies, dataset integrity, and model artifact files are configured correctly.